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Arthur Cayley

863 (concluded). Note on the Theory of Linear Differential Equations.

(Continuation of the example for $n = 2$; printed page 456.)

integer), do not involve logarithms: this is right, for the integrals are, in fact, the Legendrian functions of the first and second kinds P_p and Q_p, with only $\dfrac{1}{x}$ written therein instead of x.

Similarly, if for instance n, $n+1 = -\frac{1}{2}$, $\frac{3}{2}$, then, if $e = -n = -\frac{1}{2}$, assuming

$$x^e f \ = \ x^{-\frac{1}{2}} + Bx^{\frac{1}{2}} + Cx^{\frac{3}{2}} + Dx^{\frac{5}{2}} + Ex^{\frac{7}{2}} + \cdots ,$$

we have

$$\begin{array}{r|cccc}
 & x^{-\frac{1}{2}} & x^{\frac{1}{2}} & x^{\frac{3}{2}} & x^{\frac{5}{2}} \\
\hline
\frac{1}{2}\cdot -\frac{1}{2}\, B & -\frac{1}{4}B & & & \\
 & -\frac{3}{4}C & +\frac{3}{4}D & -\frac{7}{4}E & \\
 & +1 & -B & -3C & \\
 & -\frac{1}{2} & -\frac{3}{4}B - \frac{3}{4}C & -\frac{3}{4}D - \frac{7}{4}E & \\
P(x^e f) = & 0 & -B & 0\cdot C & 3D \\
 & & +\frac{1}{2} & -\frac{3}{4}B & 5E
\end{array}$$

We have here B if not $= 0$, $\mu = \frac{1}{2}$; but if $B = 0$, then we cannot in any way make the coefficient of $x^{\frac{1}{2}}$ to vanish, and consequently $\mu = \frac{1}{2}$. With this last value of μ, the group $(\mu, -\frac{1}{2} + n, n-1)$, that is, $(\mu, -\frac{1}{2}, \frac{3}{2})$, becomes $(\frac{1}{2}, -\frac{1}{2}, \frac{3}{2})$ which contains two equal roots, and the conclusion from the theorem thus is that the integrals u_1, u_2 corresponding to the roots $-\frac{1}{2}$, $\frac{3}{2}$ involve logarithmic values. And similarly in general the integrals u_1, u_2, corresponding to the roots $-p + \frac{1}{2}$, $p + \frac{1}{2}$ (p any positive integer), involve logarithmic values: this also is right.

The examples exhibit the true character of the theorem, and show I think that it is a less remarkable one than would at first sight appear: in fact, in working them out, we really ascertain by an actual substitution whether the differential equation can be satisfied by series of powers only, without logarithms. Thus for $n = 2$ as above, it appears that the equation is satisfied by the series

$$y = x^{-2} + Bx^{-1} + Cx^0 + Dx^1 + Ex^2 + Fx^3 + Gx^4 + \cdots ,$$

where $B = 0$, $C = -\frac{1}{2}$, $D = 0$, $E = 0$, $F = F$, $G = 0$, $H = -\frac{2}{7}F$, …, that is, by

$$y = x^{-2} + A\big(x^0 - \tfrac{1}{2}\big) + F\big(x^3 + \tfrac{1}{5}\ \cdots\big);$$

in other words, that we have the two particular integrals $y = x^{-2} + \frac{1}{2}$, and $y = x^3 + \frac{1}{5}x \cdots$, belonging to the two roots -2, 3 respectively.

Similarly, when $n = \frac{1}{2}$, we cannot satisfy the equation by a series

$$y = x^{-\frac{1}{2}} + Bx^{\frac{1}{2}} + Cx^{\frac{3}{2}} + Dx^{\frac{5}{2}} + \cdots ;$$

for in order to satisfy the equation, we must have $B = 0$, $C = \infty$, there is thus no series of powers $y = x^{-\frac{1}{2}} + Cx^{\frac{3}{2}} + \cdots$, corresponding to the root $-\frac{1}{2}$: but there is a series $y = x^{\frac{3}{2}} + kx^{\frac{5}{2}} + \cdots$ corresponding to the root $\frac{3}{2}$; and thus the integrals u_1, u_2, corresponding to these roots $-\frac{1}{2}$, $\frac{3}{2}$, involve logarithms.

Cambridge, 23 March 1887.

864. On Rudio's Inverse Centro-Surface.

[From the *Quarterly Journal of Pure and Applied Mathematics*, vol. XXII. (1887), pp. 156–158.]

DR F. RUDIO, in an inaugural dissertation "Ueber diejenigen Flächen deren Krümmungsmittelpunkts-flächen confokale Flächen zweiten Grades sind," Berlin, 1880, and *Crelle's Journal*, t. XCV., p. 240, has determined the surfaces having for their centre-surface (i.e., the locus of centres of curvature) the aggregate of the confocal quadric surfaces

$$\frac{x^2}{a-\lambda}+\frac{y^2}{b-\lambda}+\frac{z^2}{c-\lambda}=1, \qquad \frac{x^2}{a-\mu}+\frac{y^2}{b-\mu}+\frac{z^2}{c-\mu}=1,$$

or, what is the same thing, the surfaces orthotomic to the common tangents of these two surfaces. He obtains, as the final result of an elegant analytical investigation, the following formulae:

$$x=\sqrt{(a-\lambda)}\sqrt{\frac{a-u\,.\,a-v}{a-b\,.\,a-c}},$$
$$y=\sqrt{(b-\lambda)}\sqrt{\frac{b-u\,.\,b-v}{b-c\,.\,b-a}},$$
$$z=\sqrt{(c-\lambda)}\sqrt{\frac{c-u\,.\,c-v}{c-a\,.\,c-b}},$$
$$U=\sqrt{\frac{a-u\,.\,b-u\,.\,c-u}{\lambda-u\,.\,\mu-u}},$$
$$V=\sqrt{\frac{a-v\,.\,b-v\,.\,c-v}{\lambda-v\,.\,\mu-v}},$$

$$\xi\,\frac{U-V}{x}=1-\frac{b-\lambda\,.\,c-\lambda}{\lambda-u\,.\,\lambda-v}=\frac{a-\mu}{a-u\,.\,a-v}UV,$$
$$\eta\,\frac{U-V}{y}=1-\frac{c-\lambda\,.\,a-\lambda}{\lambda-u\,.\,\lambda-v}=\frac{b-\mu}{b-u\,.\,b-v}UV,$$
$$\zeta\,\frac{U-V}{z}=1-\frac{a-\lambda\,.\,b-\lambda}{\lambda-u\,.\,\lambda-v}=\frac{c-\mu}{c-u\,.\,c-v}UV,$$

(values which are such that $\xi^2+\eta^2+\zeta^2=1$),
And then

$$\rho=\tfrac{1}{2}\left(\int\frac{du}{U}+\int\frac{dv}{V}\right)+C;$$

$$x'=x+\rho\xi,\quad y'=y+\rho\eta,\quad z'=z+\rho\zeta,$$

viz. the equations give x, y, z, U, V, ξ, η, ζ, each of them as a function of two independent parameters u, v; ρ is a function of u, v and of the arbitrary constant C; hence, giving to C any assumed value, we have x', y', z' each of them a function of the two arbitrary parameters u, v; that is, x', y', z' are the coordinates of a point on a surface, one of a system of parallel surfaces (corresponding to the different values of C) which are the surfaces in question.

Observe that u, v are the elliptic coordinates of the point (x, y, z) on the first of the two confocal surfaces, and that ξ, η, ζ are the cosine-inclinations of one of the tangents from this point to the other confocal surface; so that, if ρ were left arbitrary, the equations $x'=x+\rho\xi$, $y'=y+\rho\eta$, $z'=z+\rho\zeta$ would be the equations of a common tangent of the two confocal surfaces; but ρ, as determined, is the distance of the point x, y, z from the point x', y', z' on the required surface. The expression for ρ involves hyper-elliptic integrals of the first species, which are the same as those which present themselves in the determination of the geodesic lines upon either of the two confocal surfaces.

I have, in quoting these remarkable results, written for greater simplicity a, b, c instead of the author's a^2, b^2, c^2.

Cambridge, Dec. 20, 1886.

865. On Multiple Algebra.

[From the *Quarterly Journal of Pure and Applied Mathematics*, vol. XXII. (1887), pp. 270–308.]

1. I REPRODUCE a passage from my Presidential Address, British Association, Southport, 1883.

"Outside of ordinary mathematics we have some theories which must be referred to: algebraical, geometrical, logical. It is, as in many other cases, difficult to draw the line: we do in ordinary mathematics use symbols not denoting quantities, which we nevertheless combine in the way of addition and multiplication, $a+b$ and ab, and which may be such as not to obey the commutative law $ab = ba$; in particular, this is or may be so in regard to symbols of operation; and it could hardly be said that any development whatever of the theory of such symbols of operation did not belong to ordinary algebra. But I do separate from ordinary algebra the system of multiple algebra or linear associative algebra developed in the valuable memoir by the late Benjamin Peirce, "Linear Associative Algebra" (1870, reprinted 1881 in the *American Journal of Mathematics*, vol. IV., with notes and addenda by his son, C. S. Peirce): we here consider symbols A, B, &c., which are linear functions of a determinate number of letters or units i, j, k, l, &c., with coefficients which are ordinary analytical magnitudes real or imaginary, viz. the coefficients are in general of the form $x+iy$, where i is the before-mentioned imaginary, or $\surd(-1)$ of ordinary analysis. The letters i, j, k, &c., are such that every binary combination i^2, ij, ji, &c., (ij not in general $= ji$), is equal to a linear function of the letters, but under the restriction of satisfying the associative law; viz. for each combination of three letters $ij \cdot k$ is $= i \cdot jk$, so that there is a determinate and unique product of three or more letters; or, what is the same thing, the laws of combination of the units i, j, k, ... are defined by a multiplication table giving the values of i^2, ij, ji, &c.; the original units may be replaced by linear functions of these units, so as to give rise for the units finally adopted to a multiplication table of the most simple form; and it is very remarkable how frequently in these simplified forms we have nilpotent or idempotent symbols ($i^2 = 0$ or $= i$ as the case may be), and symbols i, j, such that $ij = ji = 0$; and consequently how simple are the forms of the multiplication tables which define the several systems respectively.

"I have spoken of this multiple algebra before referring to various geometrical theories of earlier date, because I consider it as the general analytical basis, and the true basis, of these theories. I do not realise to myself directly the notions of the addition or multiplication of two lines, areas, rotations, or other geometrical, kinematical, or mechanical entities; and I would formulate a general theory as follows: consider any such entity as determined by the proper number of parameters a, b, c, ... (for instance, in the case of a finite line given in magnitude and position, these might be the length, the coordinates of one end, and the direction-cosines of the line considered as drawn from this end); and represent it by or connect it with the linear function $ai+bj+ck+$&c., formed with these parameters as coefficients and with a given set of units i, j, k, &c. Conversely, any such linear function represents an entity of the kind in question. Two given entities are represented by two linear functions; the sum of these is a like function representing an entity of the same kind, which may be regarded as the sum of the two entities; and the product of them (taken in a determined order, when the order is material) is an entity of the same kind, which may be regarded as the product (in the same order) of the two entities. We thus establish by definition the notion of the sum of the two entities, and that of the product (in a determinate order, when the order is material) of the two entities. The value of the theory in regard to any kind of entity would of course depend on the choice of a system of units i, j, k, ..., with such laws of combination as would give a geometrical or kinematical or mechanical significance to the notions of the sum and product as thus defined.

"Among the geometrical theories referred to, we have a theory (that of Argand, Warren, and Peacock) of imaginaries in plane geometry; Sir W. R. Hamilton's very valuable and important theory of quaternions; the theories developed in Grassmann's *Ausdehnungslehre*, 1844 and 1862; Clifford's theory of biquaternions, and recent extensions of Grassmann's theory to non-Euclidian space by Mr Homersham Cox. These different theories have of course been developed, not in anywise from the point of view from which I have been considering them, but from the points of view of their several authors respectively."

2. The present paper is in a great measure the development of the views contained in the foregoing extract; but, instead of establishing *ab initio* a linear function $ai+bj+ck+\cdots$ as above, I deduce this, as will be seen from the notion of addition.

3. If x, y, ... denote ordinary (real or imaginary) analytical magnitudes, which (as such) are susceptible of addition and multiplication, and for each of these operations are commutative and associative, then we may consider a multiple symbol $(x, y, \ldots)$ involving any given number of letters, to fix the ideas say

(x, y), susceptible of addition and multiplication according to determinate laws

$$(x, y) + (x', y') = (P, Q), \qquad (x, y)(x', y') = (X, Y),$$

where P, Q, X, Y are given functions of x, y, x', y'. For greater simplicity the law of addition is taken to be

$$(x, y) + (x', y') = (x + x',\ y + y'),$$

so that as regards addition the multiple symbols are commutative and associative. But this is or is not the case for multiplication, according to the form of the given functions X, Y; for instance, if

$$(x, y)(x', y') = (xx' - yy',\ xy' + yx'),$$

then in regard to multiplication the symbols will be commutative and associative. But if

$$(x, y, z)(x', y', z') = (yz' - zy',\ zx' - xz',\ xy' - yx'),$$

then the symbols will be associative, but not commutative.

4. I remark here that we are in general concerned with symbols of a given multiplicity, double symbols (x, y), triple symbols (x, y, z), n-tuple symbols $(x_1, x_2, \ldots, x_n)$, as the case may be, and that as well the product as the sum is a symbol *ejusdem generis*, and consequently of the same multiplicity, with the component symbols; this is to be assumed throughout in the absence of an express statement to the contrary. It is, moreover, proper to narrow the notion of multiplication by restricting it to the case where the terms $(X, Y, \ldots)$ of the product are linear functions of the terms $(x, y, \ldots)$ and $(x', y', \ldots)$ of the component symbols respectively; any other form $(X, Y, \ldots)$ is better designated not as a product, but as a combination (or by some other name) of the component symbols $(x, y, \ldots)$ and $(x', y', \ldots)$.

5. I assume, moreover, that, if m be any ordinary analytical magnitude, this may be multiplied into a multiple symbol $(x, y, \ldots)$, according to the law

$$m(x, y, \ldots) = (mx, my, \ldots).$$

6. As a consequence of this last assumption and of the assumed law of addition, we have for instance

$$(x, y, z) = (x, 0, 0) + (0, y, 0) + (0, 0, z) = x(1, 0, 0) + y(0, 1, 0) + z(0, 0, 1);$$

that is, using single letters i, j, k for the multiple symbols $(1, 0, 0)$, $(0, 1, 0)$, $(0, 0, 1)$ respectively, we have

$$(x, y, z) = xi + yj + zk,$$

where the letters i, j, k, thus standing for determinate multiple symbols, may be termed "*extraordinaries.*" Each extraordinary may be multiplied into any ordinary symbol x, and is commutative therewith, $xi = ix$; moreover, each extraordinary may be multiplied into itself, or into another extraordinary, according to laws which are, in fact, determined by means of the assumed law of multiplication of the original multiple symbols; and, conversely, the law of multiplication of the extraordinaries determines that of the original multiple symbols; thus, if

$$(x, y)(x', y') = (xx' - yy',\ xy' + yx'),$$

then

$$(ix + jy)(ix' + jy') = i(xx' - yy') + j(xy' + yx'),$$

and also

$$(ix + jy)(ix' + jy') = i^2 xx' + ij\, xy' + ji\, yx' + j^2 yy',$$

which expressions will agree together if, and only if, $i^2 = i$, $ij = j$, $ji = j$, $j^2 = -i$, or, as these equations may be written,

$$\begin{array}{c|cc} & i & j \\ \hline i & i & j \\ j & j & -i \end{array}$$

And so in general we have a multiplication table giving each square or product as a homogeneous linear function of all or any of the extraordinaries.

7. And, conversely, from this multiplication table of the extraordinaries i, j, we have

$$(ix+jy)(ix'+jy') = i(xx'-yy') + j(xy'+yx'),$$

that is,

$$(x,y)(x',y') = (xx'-yy',\ xy'+yx'),$$

the originally assumed formula of multiplication.

In the example just given, we have $i^2 = i$, $ij = ji = j$, viz. the symbol i comports itself like unity, and may be put $= 1$; we have $(x,y) = x + jy$, with the multiplication table

$$\begin{array}{c|cc} & 1 & j \\ \hline 1 & 1 & j \\ j & j & -1 \end{array}$$

or simply the equation $j^2 = -1$; and then

$$(x+jy)(x'+jy') = xx' - yy' + j(xy'+yx'),$$

and it is convenient to regard 1 as an extraordinary, and speak of the system of extraordinaries 1, j.

8. The separate terms x, y, …, whatever be their number, may always without loss of generality be arranged in a line; but it may be convenient to arrange them in a different form, for instance, in that of a square; we may have for instance symbols

$$\begin{vmatrix} x & y \\ z & w \end{vmatrix}$$

with the laws of combination

$$\begin{vmatrix} x & y \\ z & w \end{vmatrix} + \begin{vmatrix} x' & y' \\ z' & w' \end{vmatrix} = \begin{vmatrix} x+x' & y+y' \\ z+z' & w+w' \end{vmatrix},$$

$$\begin{vmatrix} x & y \\ z & w \end{vmatrix} \cdot \begin{vmatrix} x' & y' \\ z' & w' \end{vmatrix} = \begin{vmatrix} xx'+yz' & xy'+yw' \\ zx'+wz' & zy'+ww' \end{vmatrix},$$

where observe that the multiplication is not commutative.

9. A multiple symbol may be connected with a geometrical or physical entity of any kind: viz. any such entity, depending on a number of parameters susceptible of analytical magnitude, to fix the ideas say on two parameters (x,y), may be connected with the multiple symbol (x,y). We cannot in general directly conceive the addition or multiplication of such entities, but the assumed laws of combination of the multiple symbols in effect serve as definitions of the operations. Thus

$$(x,y) + (x',y') = (x+x',\ y+y'); \quad (x,y)\cdot(x',y') = (xx'-yy',\ xy'+yx');$$

or the sum of the entities whose parameters are x, y and x', y' is by definition the entity whose parameters are $x+x'$, $y+y'$; and the product of the same entities is by definition the entity whose parameters are $xx'-yy'$, $xy'+yx'$. The entities are thus, as regards addition, commutative and associative; but, as regards multiplication they are or are not commutative, or associative, according to the assumed law of multiplication of the multiple symbols.

10. If, as above, the multiple symbol be represented by means of extraordinaries, $(x,y) = ix + jy$, then the extraordinaries i, j, qua multiple symbols $(1,0)$, $(0,1)$, are themselves special entities of the kind in question, their laws of combination (as such special entities) being included in the general laws for the combination of such entities; or, if we please, these general laws being derived from the assumed laws for the special entities. Thus, if $ix+jy$ represent the point whose coordinates are x, y, then i represents the point whose coordinates are $(1,0)$, and j the point whose coordinates are $(0,1)$.

11. It has been assumed that the multiple symbols combined together, and their sum and product, are symbols of one and the same multiplicity, say the types of combination are

$$(x,y,z) + (x',y',z') = (x+x',\ y+y',\ z+z'); \quad (x,y,z)(x',y',z') = (X,Y,Z);$$

if this were not so, if for instance we had

$$(x,y,z)(x',y',z') = \text{a double symbol } (X,Y),$$

then this given equation for the multiplication of two triple symbols would not serve as a definition for the multiplication of double symbols, or of a double and a triple symbol, and new definitions would be required. We might, however, have symbols of indefinite multiplicity $(x, y, z, w, \ldots)$, including within them all finite multiplicities, viz. (x, y) meaning $(x, y, 0, 0, \ldots)$, and so in other cases, these being combined into symbols of like indefinite multiplicity

$$(x, y, z, \ldots) + (x', y', z', \ldots) = (x + x',\ y + y',\ z + z',\ \ldots);$$

$$(x, y, z, \ldots)(x', y', z', \ldots) = (X, Y, Z, \ldots);$$

for instance, the law of multiplication might be

$$(x, y, z, \ldots)(x', y', z', \ldots) = (xx' + yx' + zx' + \cdots,\ xy' + yy' + \cdots,\ \ldots),$$

and we could hereby combine symbols of any finite multiplicities (the same or different) whatever.

12. Another peculiarity may be noticed: suppose that, in general,

$$(x, y, z, w) + (x', y', z', w') = (x + x',\ y + y',\ z + z',\ w + w');$$

$$(x, y, z, w)(x', y', z', w') = (X, Y, Z, W);$$

then, as a particular case hereof, we have the addition-equation

$$(x, y, 0, 0) + (x', y', 0, 0) = (x + x',\ y + y',\ 0,\ 0),$$

and it may very well be that for the same two symbols the multiplication-equation may take the form

$$(x, y, 0, 0)(x', y', 0, 0) = (0, 0, Z, W),$$

(Z, W each of them a function of x, y, x', y'); viz. here, in a different point of view, the component symbols $(x, y, 0, 0)$ and $(x', y', 0, 0)$ are double systems of a certain kind $\{x, y\}$, $\{x', y'\}$, and the product is a double system of a different kind $[Z, W]$, or it might have been $(0, Y, Z, W)$, a triple system $[Y, Z, W]$. Of course, all peculiarity disappears when we revert from the particular symbols $(x, y, 0, 0)$ to the general symbols (x, y, z, w), from which we may regard them as derived.

13. The peculiarity just referred to presents itself naturally when we regard the symbols as representing geometrical or physical entities; it may very well be that the product of two entities of a certain kind is taken to be an entity of a different kind (for instance, the product of two points to be a line), and that the analytical theory of the multiple symbol is constructed in order to such a relation; and it may further be that only the symbol $(x, y, 0, 0)$ or $\{x, y\}$ is interpreted with reference to an entity of the one kind, and only the symbol $(0, 0, Z, W)$, or $[Z, W]$ is interpreted by reference to an entity of the other kind, without any interpretation at all being given to the general symbol (x, y, z, w); thus the two forms $(x, y, 0, 0)$, and $(0, 0, Z, W)$, naturally present themselves as double symbols $\{x, y\}$ and $[Z, W]$ of different kinds.

14. In further illustration, suppose the symbols represented as linear functions of extraordinaries,

$$(x, y, z, w) = xe_1 + ye_2 + z\eta_1 + w\eta_2,$$

with a multiplication table for the four extraordinaries e_1, e_2, η_1, η_2. We then have $(x, y, 0, 0) = \{x, y\} = xe_1 + ye_2$; the e_1, e_2 have no proper multiplication table of their own, but the squares and products e_1^2, e_1e_2, e_2e_1, e_2^2 are each given as a linear function of η_1, η_2; hence we have

$$(xe_1 + ye_2)(x'e_1 + y'e_2) = Z\eta_1 + W\eta_2,$$

which is a symbol $(0, 0, Z, W)$, or $[Z, W]$, of a different kind; it may, in completion of the theory, be assumed that in like manner η_1, η_2 have no proper multiplication table of their own, but that the squares and products η_1^2, $\eta_1\eta_2$, $\eta_2\eta_1$, η_2^2 are each given as a linear function of e_1, e_2, leading to a relation

$$(z\eta_1 + w\eta_2)(z'\eta_1 + w'\eta_2) = Xe_1 + Ye_2,$$

which is a symbol $(X, Y, 0, 0)$ or $\{X, Y\}$ of the first kind; it may further happen that each of the several products $e\eta$, ηe is 0, in which case

$$(xe_1 + ye_2)(z\eta_1 + w\eta_2) = 0, \quad (z\eta_1 + w\eta_2)(xe_1 + ye_2) = 0.$$

For the complete theory, we require the multiplication table of the four extraordinaries e_1, e_2, η_1, η_2. The geometrical interpretation may be that $xe_1 + ye_2$ represents a point, $z\eta_1 + w\eta_2$ a line (with or without any geometrical interpretation of a symbol $xe_1 + ye_2 + z\eta_1 + w\eta_2$), the product of two points is a line, the product of two lines is a point; that of a line and point is $= 0$, (see *post* Grassmann, where however the system is a somewhat different one).

15. I do not propose in the present paper to consider the subject of multiple algebra from an analytical point of view; and I will make only a few remarks and give some references. The general theory of associative linear forms is treated in a very satisfactory manner in Peirce's Memoir (1870) above referred to, only it is assumed throughout *ab initio* that the forms are associative. In a Note "On Associative Imaginaries," *Johns Hopkins Circulars*, No. 15 (1882), [822], and more fully in the paper "On Double Algebra," *Proc. Lond. Math. Soc.*, t. XV. (1884), pp. 185–197, [814], starting from the assumed equations $x^2 = ax + by$, $xy = cx + dy$, $yx = ex + fy$, $y^2 = gx + hy$, between the extraordinaries x, y, I considered under what conditions the algebra of these symbols was in fact associative. And in a paper "On the 8-square Imaginaries," *American Journal of Mathematics*, t. IV. (1881), pp. 293–296, [773], I showed that the extraordinaries 0, 1, 2, 3, 4, 5, 6, 7, connected with Euler's theorem of the 8 squares, are of necessity non-associative. I do not know that anything else has been done in regard to non-associative algebras: and it thus appears that the question has hardly been discussed at all. Matrices are associative: I shall have again to refer to them.

16. The object of the present paper is to discuss in detail, from the point of view explained in the extract from my British Association Address, the different theories in regard to geometrical or other entities. I do this under various headings.

The $i = \surd(-1)$ of Analysis and Analytical Geometry. Art. Nos. 17–26.

17. We may, of course, consider the i as an extraordinary. We have a double symbol (x, y), combining according to the laws

$$(x, y) + (x', y') = (x + x',\ y + y'),$$
$$(x, y)(x', y') = (xx' - yy',\ xy' + yx');$$

and then, introducing the extraordinaries k, i, and writing

$$(x, y) = kx + iy,$$

we have

$$(kx + iy)(kx' + iy') = k^2xx' + ki\,xy' + ik\,yx' + i^2yy';$$

whence

$$k^2 = k, \quad ki = ik = i, \quad i^2 = -k,$$

or the multiplication table is

	k	i
k	k	i
i	i	$-k$

But the conditions are satisfied by, and we accordingly assume, $k = 1$; and there then remains only the condition $i^2 = -1$. Compare herewith Sir W. R. Hamilton, "Theory of Conjugate Functions, or Algebraical Couples; with a Preliminary and Elementary Essay on Algebra as a Science of pure time" (*Trans. R. I. Acad.*, t. XVII., 1833–35). I refer to this paper in my Address, and remark upon it that I cannot appreciate the manner in which the author connects, with the notion of time, his algebraical couple or imaginary magnitude $a + bi$, $= a + b\surd(-1)$ as written in the memoir.

18. But we have, in fact, passed out of this view, and have come to regard $a + bi$ as an ordinary analytical magnitude; viz. in every case an ordinary symbol represents or may represent such a magnitude, and the magnitude (and as a particular case thereof, the symbol i) is commutable with the extraordinaries of any system of multiple algebra. And similarly in analytical geometry, without seeking for any real representation, we deal with imaginary points, lines, &c., that is, with points, lines, &c., depending on parameters of the foregoing form $a + bi$.

$\surd(-1)$ denotes Perpendicularity. Art. Nos. 19–26.

19. I give a list of the earlier works and memoirs, marking with an asterisk those which I have not seen.

WALLIS. *De Algebra Tractatus*, 1685 Anglice editus; 1693. Chapters 66, 67, 68, and 69 have respectively the titles: *De quadratis negativis eorumque radicibus dictis Imaginariis; Eorundem exemplificatio in Geometria; Effectiones geometricæ his accommodatæ; Aliæ quæ huc spectant constructiones geometricæ.* I quote a single sentence from Chap. 67: "Prout igitur cum æquationis quadraticæ radix prodit negativa, dicendum verbi gratia punctum B pro eo statu haberi non posse ut supponitur in AC exposita prorsum; posse tamen retrorsum ab A in eadem recta; hic vero (de radice quadrati negativi) dicendum, non haberi quidem posse punctum B ut erat suppositum in AC recta, vel ante vel retro: posse tamen (in eodem plano) supra rectam illam": viz. the distance represented by the square root of a negative quantity cannot be measured in the line, forwards or backwards; but can be measured (in the same plane) above the line or, as appears elsewhere, at right angles to the line, either in the plane, or in a plane at right angles thereto.

FORCENEX. "Réflexions sur les quantités imaginaires," *Mél. de Turin*, t. I. (1759), pp. 113–146. The author, after stating that there was no geometrical construction for imaginary quantities, proceeds "Cependant, pour conserver une certaine analogie avec les quantités négatives, un auteur dont nous avons un cours d'algèbre d'ailleurs fort estimable a prétendu les devoir prendre sur une ligne perpendiculaire à celle où l'on les avait supposées, si par exemple &c.," the example which he considers being as follows: On a given line AB to find a point P, such that the product of the distances AP, BP is $= \frac{1}{2}(AB)^2$. Taking $AB = 2a$ and $AP = x$, then $x(2a - x) = 2a^2$, that is, $(x - a)^2 = -a^2$, or $x = a + a\sqrt{(-1)}$, an imaginary value; the condition is, however, satisfied by a real point P off the line AB

$$(AM = MP = \tfrac{1}{2}\,AB = a), \quad \text{(fig. 1)},$$

and hence the author in question infers that the imaginary value $a\sqrt{(-1)}$ is represented by the line MP at right angles to AB. Forcenex objects to this: if the

[Figure 1: line AB with midpoint M, and point P directly above M at distance a, forming a right angle with AB.]

point P may be taken off the axis of x, then the condition $AP \cdot BP = 2a^2$ gives for the locus of P a certain curve; and there is no reason why the foregoing solution $x = a + a\sqrt{(-1)}$ should represent one point rather than another of this curve. As to this see *post* (Peacock).

*TRUEL, Henri Dominique, 1786.

*SUREMAIN-DE-MISSERY. *Théorie purement algébrique des quantités imaginaires*, Paris, 1801; it is referred to in the letter by Servois, *infra*.

20. BUÉE. "Mémoire sur les quantités imaginaires," *Phil. Trans.*, 1806, pp. 23–88.

I give an extract: "Du signe $\sqrt{(-1)}$. Je mets en titre, du signe $\sqrt{(-1)}$ et non de la quantité imaginaire ou de l'unité imaginaire, parce que $\sqrt{(-1)}$ est un signe particulier joint à l'unité réelle, non une quantité particulière: c'est un nouvel adjectif joint au substantif ordinaire et non un nouveau substantif.

"Mais que veut dire ce signe? il n'indique ni une addition ni une soustraction, ni une suppression ni une opposition par rapport aux signes $+$ et $-$: une quantité accompagnée par $\sqrt{(-1)}$ n'est ni additive ni subtractive, ni égale à zéro. La qualité marquée par $\sqrt{(-1)}$ n'est opposée ni à celle qu'indique $+$, ni à celle marquée par $-$: qu'est-elle donc?

"Pour la découvrir supposons trois lignes égales AB, AC, AD (fig. 2) qui partent toutes du point A. Si je désigne la ligne AB par $+1$, la ligne AC sera -1, et la ligne AD qui est une moyenne proportionnelle entre $+1$ et -1, sera nécessairement $\sqrt{(-1 \cdot a)}$ ou plus simplement $\sqrt{(-1)}$. Ainsi $\sqrt{(-1)}$ est le signe de la PERPENDICULARITÉ: donc la propriété caractéristique est que tous les points de la perpendiculaire sont également éloignés de points placés à égale distance de part et d'autre de son pied. Le signe $\sqrt{(-1)}$ signifie tout cela et il est le seul qui l'exprime.

[Figure 2: three equal lines AB, AC, AD from point A, with C to the left of A on the horizontal through B, and D vertically above A.]

"Ce signe mis devant a (a signifiant une ligne ou une surface) veut donc dire *qu'il faut donner à a une situation perpendiculaire à celle qu'on lui donnerait si l'on avait simplement a ou $-a$.*"

And he, in fact, gives (Prob. vi.) the problem before referred to, and finds no difficulty in assigning to P a position off the line AB. I notice also that Buée considers a curve as having branches in a perpendicular plane; thus for the circle $x^2 + y^2 = a^2$ in the plane of the paper, putting $-x^2$ in place of x^2, there are branches, or, as he calls it, "une appendice," if $y^2 = a^2 + x^2$ (or say $y^2 = a^2 - z^2$), being a rectangular hyperbola, in a plane at right angles to the plane of the paper.

21. Other writings are:

ARGAND. *Essai sur une manière de représenter les quantités imaginaires.* Paris, 1806. Partly reproduced in Argand's memoir under the same title presently referred to.

FRANÇAIS, J. F. "Nouveaux principes de géométrie de position et interprétation géométrique des symboles imaginaires." *Gergonne Annales*, t. IV. 1813–14, pp. 61–72.

ARGAND. "Essai sur une manière, &c." *Ibid.* pp. 133–148.

FRANÇAIS, J. F. "Lettre au Rédacteur." *Ibid.* pp. 222–227.

SERVOIS. "Lettre au Rédacteur." *Ibid.* pp. 228–235.

This last is against the theory, with a running commentary of notes in favour by Gergonne.

FRANÇAIS, J. F. "Autre lettre." *Ibid.* pp. 364–366.

LACROIX. "Note transmise à M. Vecten"; it calls attention to Buée's memoir. *Ibid.* p. 367.

*MOUREY. *Vraie théorie des quantités négatives et des quantités prétendues imaginaires*, 1828, reprinted 1861.

22. WARREN, J. *A treatise on the geometrical representation of the square roots of negative quantities.* 8vo. Cambridge, 1828, pp. 1–154.

PEACOCK, Preface to *Algebra*, 1830, speaks of Warren's work as "distinguished for great originality and for extreme boldness in the use of definitions." There is no Preface or Introduction; the first Chapter is entitled Definitions, Addition, Subtraction, Proportion, Multiplication, Division, Fractions, and Raising of Powers. I quote certain articles as follows:

(1) All straight lines drawn in a given direction from a given point are represented in length and direction by algebraic quantities; and in the following treatise whenever the word quantity is used it is to be understood as signifying a line.

(2) DEF. The given point from which the straight lines are measured is called the origin.

(3) DEF. The sum of two quantities is the diagonal of the parallelogram whose sides are the two quantities.

(12) DEF. The first of four quantities is said to have to the second the same ratio which the third has to the fourth: when the first has in length to the second the same ratio which the third has in length to the fourth, according to Euclid's definition; and also the angle at which the fourth is inclined to the third is equal to the angle at which the second is inclined to the first, and is measured in the same direction.

(17) DEF. Unity is a positive quantity arbitrarily assumed, from a comparison of which the values of other quantities are obtained.

(18) DEF. If there be three quantities such that unity is to the first as the second is to the third, then the third is called the product which arises from the multiplication of the second by the first.

23. The signification of these definitions may be thus expressed. Let the line, drawn from an origin O to the point whose rectangular coordinates are x, y, be called the line $x + iy$; of course the line, length unity in the direction of the axis of x, is the line 1. Then, observing that, putting x, $y = r\cos\theta$, $r\sin\theta$, and x', $y' = r'\cos\theta'$, $r'\sin\theta'$, we have $(x+iy)(x'+iy') = rr'\{\cos(\theta+\theta') + i\sin(\theta+\theta')\}$, the two definitions are:

The sum of the lines $x + iy$ and $x' + iy'$ is the line

$$x + x' + i(y + y');$$

and the product of the same lines is the line

$$(x + iy)(x' + iy');$$

viz. Warren in effect establishes by definition the notions of the sum and product of two lines, and that by connecting the line with a linear symbol $x + iy$.

It may be right to remark that the general notion of a proportion, used as above to define multiplication, is geometrically the more simple of the two, and that the condition for the proportion $OA : OB = OC : OD$

may be thus expressed: the lengths of the lines are proportional, and the inclination OA to OB is equal to the inclination OC to OD, these inclinations being in the same sense.

24. PEACOCK, G. *A Treatise on Algebra.* 8vo. Cambridge, 1830 (one Vol., pp. 5–38 and 1–685).

"Report on the Recent Progress and Present State of Certain Branches of Analysis." *B. A. Report*, (1833), pp. 185–352.

A Treatise on Algebra. Vol. I. Arithmetical Algebra; Vol. II. Symbolical Algebra and its Application to the Geometry of Position. 8vo. Cambridge, 1842 and 1845.

The statement of the theory is substantially identical with that of the earlier writers, thus *Algebra*, (1830), p. 362, No. 439, is: "We have shown on a former occasion that, if a designated a line in one direction, $-a$ must designate a line in the direction opposite to it, or making an angle with the former equal to two right angles or 180 degrees; but inasmuch as $a\{\sqrt{(-1)}\}^2 = -a$, it follows that the double affectation of the line a with the sign $\sqrt{(-1)}$ produces a result represented by $-a$, and is consequently equivalent to its transfer through 180°: it follows therefore that its single affectation with the sign $\sqrt{(-1)}$ is equivalent to its transfer through half that angle, or through 90°; or, in other words, if a represents a line, $a\sqrt{(-1)}$ will represent a line at right angles to it." But further, pp. 666, 667, the Author considers the before-mentioned problem (for convenience, I write $2a$ instead of a) in the form "to divide a line $2a$ into two such parts that the rectangle contained by them may be $= b^2$"; he takes x, y for the two parts, and in the case where $b^2 > a^2$ finds the two parts

$$x = a - \sqrt{(-1)}\sqrt{(b^2 - a^2)}, \quad y = a + \sqrt{(-1)}\sqrt{(b^2 - a^2)},$$

which he interprets as meaning the two parts are AC, CB, where C is a point off the line AB.

25. It seems to me that, except in Warren, the defect in the exposition of the theory is that it is not made clear, that the theory is in fact a theory of lines in a plane through a given point, with assumed definitions for the addition and for the multiplication of lines; thus in Peacock's problem to "divide a line a into two such parts that the product of them is $= b^2$," meaning the real "given a line OA, $= a$, and a line OB, $= b^2$ (a and b^2 real and positive, so that each of these lines is in the given direction Ox), to find lines OP, OQ such that their sum may be equal to the given line OA, and their product equal to the given line OB." To solve it, take the two lines to be $x + iy = z$, and $x' + iy' = z'$; then, working with z and z', we have $z + z' = 2a$, $zz' = b^2$, giving a solution which may be written in the two forms

$$\{z = a + \sqrt{(a^2 - b^2)}, \quad z' = a - \sqrt{(a^2 - b^2)}\},$$

and

$$\{z = a + i\sqrt{(b^2 - a^2)}, \quad z' = a - i\sqrt{(b^2 - a^2)}\},$$

viz. for $a > b$, the two lines lie each of them in the line Ox, while for $b > a$, they are inclined at equal angles on opposite sides of the line Ox. Or if we work with the real values x, y, x', y', then the same two equations give $x + x' = 2a$, $y + y' = 0$, $xx' - yy' = b^2$, $xy' + yx' = 0$; viz. writing $y' = -y$, we have $x + x' = 2a$, $xx' + y^2 = b^2$, $y(x' - x) = 0$, whence either $x' = x$ or else $y = 0$, and we have the same two solutions as before. Observe that the second condition is $zz' = b^2$, not (as Forcenex understood it) the product of the lengths OP, $OQ = b^2$; and his objection is thus answered. *[Discussion continues on the next page.]*